

Muftawo Omar

Senior AI and Data Engineer | Applied AI | AI Systems | Enterprise AI | LLM Evaluation and Platforms

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PROFESSIONAL SUMMARY

Senior AI and data professional with experience spanning enterprise AI strategy, LLM benchmarking and evaluation, AI-enabled automation, data engineering, machine learning, computer vision, and cloud analytics. Builds practical systems across Python, SQL, cloud platforms, Docker, and modern AI tooling; experienced in RLHF, fine-tuning, prompt design, tool calling, multi-agent workflows, and reproducible evaluation.

TECHNICAL SKILLS

AI and LLM: OpenAI APIs and Agents SDK, Anthropic API and tool use, MCP, LangChain, LangGraph, LlamaIndex, PydanticAI, Hugging Face Transformers, vLLM, LiteLLM, LLM evaluation, prompt engineering, RLHF, fine-tuning, AI agents, multi-agent systems, RAG, function and tool calling, Power Automate, Azure AI Foundry.

Data and ML: Python, SQL, PySpark, Spark, Databricks, Kafka, Airflow, Prefect, dbt, DuckDB, Polars, Pandas, NumPy, scikit-learn, MLflow, Weights & Biases, Feast, Ray, BentoML, computer vision, simulation, signal processing.

Platforms and Production: AWS (S3, Redshift, Glue, RDS, EC2, QuickSight), Azure, Docker, Kubernetes, Terraform, CI/CD, FastAPI, Pydantic, PostgreSQL, MySQL, Redis, MongoDB, Snowflake, DynamoDB, Prometheus, Grafana.

Retrieval and Automation: pgvector, Qdrant, Weaviate, Pinecone, FAISS, Microsoft Power Automate, n8n, Microsoft Graph API, Power BI, Tableau, Dash Plotly, Streamlit, Selenium.

PROFESSIONAL EXPERIENCE

Senior AI and Data Associate | [Genser Energy](#)

July 2026 - Present

US-based energy company with operations across Africa.

- Designed and built end-to-end meter and power-reading automation across power plants' SCADA systems, applying AI and data workflows to operational processes
- Built an enterprise AI usage analytics pipeline across 500+ Cloud Enterprise seats, identifying inefficient token, context-window, model, and agent usage patterns to reduce AI infrastructure costs.
- Designed an agent indexing approach for large workbooks, enabling agents to retrieve relevant information instead of repeatedly loading entire workbook contexts; reduced a high-usage user's average token spend by ~85% while improving workflow performance.
- Initiated and led an organization-wide AI adoption training on efficient AI use and optimal setups, moving employees from chat-heavy AI usage to making use of projects, context, and skills for optimal workflows.
- Conducted and developed finance, compliance, and reporting AI use-case discovery with business stakeholders; shaped practical paths from opportunity to implementation.

Senior AI Engineer, [Turing](#)

July 2025 - Aug 2026

Turing is an IT talent services provider for Global enterprises

Lead AI Engineer - (Meta, Inc.) - Multi-Agentic Coding Annotator

- Designed and built high-quality multi-agent benchmark tasks based on real-world software engineering workflows.
- Wrote clear, precise task instructions specifying file paths, function signatures, expected behavior, and constraints while employing the Harbor evaluation framework to run and validate tasks inside Docker environments
- Created decomposition instruction strategies that split complex code changes across multiple independent sub-agents

AI Engineer - (Meta, Inc.)

- Fine-tuned and evaluated prereleased LLaMA models across multiple coding tasks and benchmarked results using predefined performance rubrics.
- Designed few-shot prompting strategies tailored to pretrained LLaMA models, enabling consistent and alignment-free evaluation across programming benchmarks.
- Developed contamination-resistant prompts and deterministic test protocols to ensure fair and reproducible evaluation of foundational LLaMA models on unseen coding tasks.

Lead AI Trainer (LLM)- (Microsoft Inc.)

- I spearhead training multiple Large Language Models (LLMs) using Reinforcement Learning with Human Feedback (RLHF), specializing in programming languages, software engineering concepts, and advanced programming paradigms.
- Conduct detailed evaluations of LLMs, fine-tuning their performance to ensure optimal accuracy and relevance

in real-world applications.

Data Engineer - [Kofa Technologies](#)

Jan 2022 - June 2025

IoT-based Battery Network Management and E-mobility company; founding data scientist and sole data-team member.

Senior Data Engineer/Scientist

- Pricing Model Development: Formulated comprehensive pricing models for motorcycle and battery access that enhance transparency, improving customer satisfaction and savings by approximately 15%.
- End-to-end Data Pipelines: Built data pipelines and dashboards leveraging AWS services, leading to a 100% adoption of data-driven decision-making across departments.
- Location Optimization: Developed an unsupervised machine learning model to identify optimal swap station locations, facilitating successful network expansion in Accra, Ghana.
- Simulation Modeling: Developed an object-oriented discrete event simulation model using the Python package Salabim to simulate Kofa's battery network dynamics, informing the strategic, financial, and operational decisions. This model was crucial in informing the number of batteries and stations installed in the initial phase.

>>Here in a [Joy News \(Ghana\) Air Quality bulletin](#) at 3:50, I demonstrated how we utilized dashboards(AWS QuickSight) to track CO2 emissions.

Data Scientist - [Amalitech Services GmbH](#)

Jan 2020 – Jan 2022.

Amalitech Services GmbH is an IT talent services provider for Global enterprises.

Senior Data Scientist

>> Project Name: Aegis Rider (Advanced AI Rider Assistance Guardian System)

- Led an agile data team with experts from ETH-Zurich and Ghana, leveraging computer vision, robotics, and digital mapping to develop a rider assistance system that enhances motorbike riding safety and experience
- I implemented a signal processing pipeline that aligned data streams from different sensors, ensuring the sensor recordings were aligned for data quality purposes.

Staff Data Scientist

- I trained computer vision models that learned to predict speed, yaw, pitch, and roll from ride images.
- I implemented a signal processing algorithm that aligned data streams from different sensors, ensuring sensor recordings are aligned for data quality purposes.
- Led the annotation of a LiDAR 3D point cloud dataset to identify objects using SusTech, which was used to train and object a LiDAR object detection model.

Trainer Assistant (Contract) - NBU Hub

Feb 2019 - Dec 2019

- I was primarily responsible for assisting Youth Coaches and Trainers at the Hub, checking trainee attendance, and tracking their progress.
- I assisted trainees in resume development and pitch deck preparation.
- I taught technical skills to trainees at the NBU Hub, including Microsoft Office Suite and computer programming.

Data Analyst (Contract) - Ghana National Service Scheme

Sep 2017 - Nov 2018

- Used Python modules like pandas, numpy, and matplotlib to clean, analyze, and generate insights from data.
- Performed data analysis and cleaning in Excel.
- Developed dashboards and data visualizations with Tableau.

EDUCATION

Micro Masters in Statistics and Data Science, MITx on edX
BSc. Actuarial Science, University for Development Studies

Feb 2020 — Dec 2020
Sep 2013 — Jul 2017

SIDE PROJECTS

- [Coulkoo](#) is an LSH-based image deduplication tool for dataset preparation for AI.
- [Bitcoin Lightning Network Stats Dataset Builder](#).
- [GitMetrica](#) is an organizational GitHub activity insights tool written in Python.
- [EIT Dashboard](#) is a Streamlit dashboard for the EIT dataset.

CERTIFICATES

- Micromasters in Statistics and Data Science, Massachusetts Institute of Technology (edx.org) [certificate](#)
- Introduction to Computer Science, Massachusetts Institute of Technology (edx.org) [certificate](#)
- Data Visualization with Tableau, University of California, Davis (coursera.org) [certificate](#)
- AI Programming with Python Nanodegree (udacity.com) [certificate](#)